



FACULTY
OF LAW



UNIVERSITY OF GOTHENBURG
SCHOOL OF BUSINESS, ECONOMICS AND LAW

Carbon Capture & Storage

at Sea

Book of Abstracts

7–8 May 2026

Lund, Sweden

About the Conference

The Carbon Capture and Storage at Sea 2026 conference brings together legal scholars, policymakers, and industry professionals to examine the emerging governance frameworks for offshore CO₂ storage. This book contains abstracts from keynote speakers and presentations across five thematic panels, addressing the regulatory, legal, and environmental challenges of CCS deployment in marine environments.

Organisers

Lund University, Faculty of Law

University of Gothenburg, School of Business, Economics and Law

CONTENTS

Day 1 — Thursday, 7 May 2026

Keynote Addresses

C. Banet — Ensuring Good Resource Management	6
G. Argüello & O. Bokareva — The Legal Nature of CO ₂	8

Panel 01 · Ocean Governance & Sub-Seabed Storage

C. Skahan — Cross-Border Governance & Public Trust	10
T. Cabus — In Situ CO ₂ Mineralisation Beyond National Jurisdiction	11
V. Rafaly — Due Diligence & Time in the Law of the Sea	12
M. McElhinney — Cross-Border Carbon Transport & Storage	13
A. Wee — Justice in Carbon Sequestration at Sea	14
S. Bhattacharya & B. K. Sachdev — UNCLOS & EU Law	15

Panel 02 · Law of the Sea & CO₂ Transport

S. Mamini — Governing CO ₂ in Shipping	17
L. Fernandes Coelho — BBNJ Agreement & CO ₂ Transportation	18
E. Engebretsen — Norwegian CCS Logistics Modelling	19
L. Bilgili — Transboundary CO ₂ Shipping Challenges	20
A. Kaur & R. Dhaliwal — Maritime Liability & Liquefied CO ₂	21

Panel 03 · Regulatory Frameworks & Legal Standardisation

I. Ombudstvedt — Technical Standardisation in CCS	23
L. Østgaard — Norway's First-Mover CCS Projects	24
C. E. Meier — The EU CCS Policy Landscape	25
H. von Zweigbergk & A. Månsson — Simplification of Rules	26
N. Carnbrand Håkansson — Swedish Energy Policy & CCS	27

Day 2 — Friday, 8 May 2026

Hybrid Sessions

M. Querel — The HNS Convention & CO ₂ Transport	29
R. Sukesan — Transboundary Maritime Transport of CO ₂	30

A. Pury & P. Chaturvedi — Recalibrating Maritime Liability	31
A.-A. Bouramdane — CCS in the Nordic Context	32
M. N. A. Khan & B. Dhoundiyal — Sub-Seabed Storage in ABNJ	33
B. E. Klerk — ITLOS and ICJ Climate Opinions	34
H. Mwesigwa — CCS as Shield or Double-Edged Sword	35

Keynote Address

N. Krabbe — Biological Carbon Sequestration vs CCS	37
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Panel 04 · Civil Liability, Insurance & Risk

R. Manzo — Climate Litigation & Due Diligence in CCS	39
R. A. Bove — From Operator to State Liability	40
A. Ladynenko — Insurance Market Challenges	41

Panel 05 · National Legislation & Cross-Border Governance

D. Somoni — Emerging Regulatory Strategies	43
P. Seroogy — Precautionary Principle & the Baltic Sea	44
C. Lasselin — Marine CCS & Basaltic Rocks	45
B. A. Alexandersdóttir — Iceland Permitting Challenges	46
J. L. Díez i Besora — Port Adaptability for CCS	47
M. Krzykowski & B. Prusik — Polish CCS Legislation	48



Keynote Address

09:30–09:50 · Telaris

Catherine Banet

University of Oslo, Norway

Ensuring Good Resource Management When Developing Offshore CO₂ Storage Activities: An Appraisal of Available Regulatory Tools

Catherine Banet

University of Oslo, Norway

Good resource management has been and remains a central tenet of Norwegian petroleum policy. It is multifaceted and is reflected in many provisions of the currently applicable 1996 Petroleum Activities Act. Good resource management is also referred to by the authorities in relation to CCS because the value chains of petroleum extraction and transportation/storage of CO₂ have multiple similarities. The similarities in the operations and the legal regimes could allow for more interaction between the two sectors. The presentation discusses which regulatory instruments can contribute to good resource management of CO₂ storage capacity, reviewing the EU Net Zero Industry Act and national legislation in Norway. The main research questions are: Do we have the required legal and regulatory framework to appraise CO₂ storage as a resource and ensure good resource management? Do we have the sufficient coordination tools at national, regional and EU/EEA level?



Keynote Address

09:50–10:10 · Talaris

Gabriela Argüello & Olena Bokareva

University of Gothenburg and Lund University, Sweden

The Legal Nature of CO₂

Gabriela Argüello and Olena Bokareva

University of Gothenburg and Lund University, Sweden

Invited presentation — abstract not submitted.



Ocean Governance & the Law of Sub-Seabed Storage

10:20–12:20 · Telaris

Cody Skahan

Tony Cabus

Vonintsoa Rafaly

Michael J. T. McElhinney

Amanda Wee

Sumanta Bhattacharya & Bhavneet Kaur Sachdev

After the Pilot: Building Governance and Public Trust in Cross-Border Carbon Storage

Cody Skahan

University of Oxford, United Kingdom

Carbfix's proposed Coda Terminal is a novel cross-border carbon capture and storage (CCS) project that aims to import up to three million tonnes of CO₂ annually to Iceland for onshore mineral storage. While distinct projects, perceptions of the now-defunct Running Tide project—a marine carbon dioxide removal (mCDR) startup aimed to sequester carbon in the ocean through ocean alkalinity enhancement and algae production--has affected the regulatory and political environment for CCS projects like Coda. Both Carbfix and Running Tide were first-of-their-kind projects in Iceland, requiring the development of bespoke regulatory and permitting frameworks. Carbfix has played a key role in fostering the development of cross-border CO₂ transport and the place of mineralization in EU ETS schemes--but proper governance and policy mechanisms for Iceland to utilize CCUS to help contribute to its greenhouse gas emission reduction targets are lacking. These oversights are worrisome given that Running Tide has already been widely perceived as circumventing regulatory requirements—an approach that ultimately backfired and damaged confidence in offshore and transboundary CCS/CDR initiatives among critical experts, politicians, and the public. The Coda Terminal project has also experienced its own skepticism and criticisms--partially as a result of Running Tide, but also Carbfix's own oversights. Drawing on anthropological fieldwork conducted in Iceland and abroad with regulators, policymakers, and other actors involved in cross-border carbon transport and storage at sea, this paper examines how these two projects have shaped perceptions of environmental governance and political integrity in emerging CCS and CDR regimes. It argues that positive public attitude—alongside formal regulatory compliance—is critical for securing social license and policy continuity. Poorly governed pilot projects risk triggering backlash that may constrain future CCS deployment, and may fail to incorporate CCS into achieving EU climate targets that depend on the rapid scaling of cross-border carbon management infrastructures.

The Legal Governance of In Situ CO₂ Mineralisation Beyond National Jurisdiction

Tony Cabus

Walther-Schücking Institute for International Law, Kiel University, Germany

Carbon Capture and Storage (CCS) has attracted significant attention as a climate mitigation strategy. However, CO₂ mineralisation — the process of converting dissolved CO₂ into stable carbonate minerals within mafic and ultramafic rock formations — has largely been overshadowed by the more straightforward practice of injecting supercritical CO₂ streams into depleted oil and gas reservoirs. This disparity in popularity should not be understood as reflecting a qualitative difference between the techniques, but rather the greater technological maturity of conventional geological storage. Nonetheless, CO₂ mineralisation is gaining momentum, with several projects demonstrating promising results and distinctive advantages, particularly in terms of long-term storage security. As with climate mitigation technologies more generally, mineralisation should not be regarded as a substitute for existing approaches, but rather as a complementary option within a broader portfolio of solutions. For actors that could particularly benefit from this technique, it is therefore important to examine the conditions under which mineralisation could be deployed. To date, most projects have focused on technical feasibility and have remained within areas under national jurisdiction, thereby avoiding complex governance questions. Should demand for mineralisation increase, however, States may begin to consider sites beyond national jurisdiction. Since oceanic ridges suitable for mineralisation extend into areas beyond national jurisdiction, some States might seek to conduct operations there. Such developments would raise significant legal and governance challenges for the international seabed regime and would need to be assessed within the framework of the United Nations Convention on the Law of the Sea (UNCLOS). This paper explores these challenges, with particular attention to the Common Heritage of Mankind principle and the implications of the recently adopted Agreement on the Conservation and Sustainable Use of Marine Biological Diversity of Areas Beyond National Jurisdiction (BBNJ Agreement).

Due Diligence in Carbon Storage: An Appraisal of Time in the Law of the Sea

Vonintsoa Rafaly

World Maritime University — Sasakawa Global Ocean Institute, Sweden

Time has complex and multifaceted aspects in international affairs. This is so with regard to environmental matters, for instance, in addressing future environmental damage to the marine environment. Legal tools exist to address time-related matters, primarily through due diligence, the precautionary approach (or principle), and environmental impact assessment. Yet questions remain on the extent of the temporal scale within which such tools could operate. Due diligence is core to risk management and liability for activities that pose continuous risks to the marine environment. As Besson rightly pointed out, the temporal scope of due diligence is based on "the rights and interests to protect and the duty-bearers controlling the sources of the (risk of) harm to these rights and interests." Drawing on these two elements, this study will assess the temporal scale of due diligence in the law of the sea and its applicability to carbon storage activities. Suppose that at the end of the century, a carbon storage in areas beyond national jurisdiction has leaked and severely impacted marine biodiversity and environment, how would due diligence play out to address risk management and liability?

Regulation of Cross-Border Carbon Transport and Storage: The Role of Bilateral and Multilateral Agreements

Michael J. T. McElhinney

University of Bergen — NorthCarb Project, Norway

This presentation focuses on transboundary storage, its legal definition in different scenarios, applicable instruments, and regulatory pathways for management in the North Sea. Two principal questions are addressed: first, how to define transboundary storage and the legal considerations, exploring the extent to which the law should encapsulate different CCS operations under the concept; and second, how cross-border governing agreements for storage can be formulated, drawing comparisons with existing unitisation agreements for straddling oil and gas fields in the North Sea. In analysing these questions, consideration is given to issues of carbon plume migration and pressure propagation and their effects on both leakage risk and capacity limitations in a shared reservoir—key factors in assessing how to uphold legal duties of transboundary cooperation and equitable utilisation of shared resources. The presentation also considers liability in transboundary storage projects from both private and public perspectives.

Implications of Justice on the Evolving Law Governing Carbon Sequestration at Sea

Amanda Wee

NUS Centre for International Law, Singapore

Governance of carbon sequestration at sea remains deficient and fragmented despite its increasing potential as a tool to mitigate climate change by permanently sequestering large amounts of CO₂ at sea. This paper argues that the existing problems with the law emanate from the lack of firm undergirding principles, and that principles of justice should be incorporated into the legal framework. Such principles—including fairness, intergenerational equity, ecological responsibility, and distributive justice—will illuminate the gaps in existing law and guide its evolution in a just and equitable manner, considering the circumstances of developing States and the reliance of vulnerable communities and small island developing States on the oceans. Using these principles, the paper highlights issues of unequal distribution of risks and benefits among States and generations, transboundary pollution, uncertain application of laws, and uncertain long-term liability, compensation, and protection of the marine environment under the London Convention and its 1996 Protocol. The paper argues that a principled deliberation rooted in justice will promote the fair distribution of burdens and benefits and robustly engage with the long-term trajectory and uncertain risks of carbon sequestration at sea.

Governing CCS at Sea under UNCLOS and EU Law: Harmonizing Legal Frameworks for Sustainable Carbon Sequestration

Sumanta Bhattacharya and Bhavneet Kaur Sachdev

Asian International University, Manipur, India

Offshore Carbon Capture and Storage (CCS) is gaining attention as a key element in achieving mitigation targets for climate change. This paper analyses the governance of offshore CO₂ storage under international and regional law, specifically UNCLOS and EU law in order to coordinate transboundary sequestration, sub-seabed storage, and liability regimes. The analysis homes in on three primary problems. Firstly, the transport of CO₂ across maritime borders leads to jurisdictional challenges that existing UNCLOS rules and EU directives inadequately resolve. Second, sub-seabed sequestration has environmental and safety concerns that create lacunae between international monitoring, reporting, and verification (MRV) provisions and those under the European Union. Third, frameworks for liability in case of leakage or damage to the environment remain piecemeal. The article pleads for coherence of governance integrating UNCLOS obligations and European law instruments, with key recommendations including common MRV procedures, well-defined cross-border competency, and full liability regimes balanced between ecological preservation and technological applicability.

02

Law of the Sea & CO₂ Transport

13:20–15:00 · Alfa Laval Agri

Simone Mamini

Luciana Fernandes Coelho

Erna Engebretsen

Levent Bilgili

Akrita Kaur & Rukhsaar Dhaliwal

Governing CO₂ in Shipping: Strengthening the IMO–UNCLOS Framework

Simone Mamini

City St George's, University of London, United Kingdom

The rapid emergence of CO₂ ship transport and sub-seabed sequestration as essential components of global decarbonisation strategies exposes significant regulatory gaps in international law. As offshore storage becomes indispensable and CO₂ shipping expands across borders, unresolved questions of liability, marine environmental protection, and jurisdiction highlight the inadequacy of the current fragmented framework, which relies on the London Protocol, UNCLOS, and MARPOL. Addressing these challenges requires a coherent and legally binding international architecture capable of governing CCS activities at sea. It is argued that strengthening the IMO's GHG framework, grounded in a harmonised interpretation of UNCLOS and MARPOL, is a prerequisite for the safe, effective, and equitable development of CO₂ transport and offshore storage. The recent ITLOS Advisory Opinion on Climate Change provides a pivotal legal foundation: by confirming that anthropogenic GHG emissions constitute marine pollution under UNCLOS, the Tribunal activates binding obligations on States to prevent, reduce, and control such emissions and to cooperate in developing necessary international rules and standards. Building on this jurisprudential guidance, it is demonstrated that the current reliance on economic instruments within the IMO GHG Strategy alone cannot support the CCS sector, and that a unified legal framework integrating ITLOS-informed obligations into the IMO regulatory process is the most credible pathway for governing both CO₂ shipping and maritime decarbonisation.

Implications of the BBNJ Agreement on Transboundary CO₂ Transportation by Ships

Luciana Fernandes Coelho

Lund University, Sweden

Amidst the imperative for climate mitigation, CCS is increasingly conceptualised as a critical technology for decarbonising hard-to-abate sectors. Offshore CCS—comprising the capture, maritime transport, and sub-seabed sequestration of CO₂—operates at the intersection of fragmented jurisdictional and sectoral regimes, dominated by the 1972 London Convention and its 1996 Protocol. However, the adoption of the Agreement on Marine Biological Diversity of Areas Beyond National Jurisdiction (BBNJ Agreement) introduces a significant new layer of governance that complicates this existing framework. This paper critically evaluates the interplay between the BBNJ Agreement and the framework governing transboundary CO₂ transportation by ships, focusing on the “not undermining” mandate enshrined in Article 5.2. Specifically, it examines how BBNJ provisions on Area-Based Management Tools may impose supplementary safety obligations to mitigate CO₂ leakage in ecologically sensitive regions, and how its Environmental Impact Assessment requirements apply to CCS activities within national jurisdictions that may cause substantial pollution to the marine environment in areas beyond national jurisdiction. Central to this inquiry is whether the BBNJ Agreement functions as a complementary regime or a normative constraint. By identifying potential jurisdictional overlaps and governance gaps, the paper argues for a collaborative interpretation of Article 5.2, essential to ensure that the promotion of climate mitigation via sub-seabed sequestration does not erode the conservation of marine biodiversity in ABNJ.

Logistics Modelling of Norwegian CCS Network: Trade-Offs Between Efficiency and Environmental Integrity

Erna Engebretsen

BI Norwegian Business School, Norway

Large-scale initiatives such as Norway's Northern Lights project illustrate a rapidly evolving landscape for ship-based CO₂ transport and offshore geological storage, raising critical questions about the governance, regulatory design, and operational viability of future CCS networks. This study develops a logistics optimisation framework to evaluate how national-scale CCS systems can be configured to balance economic efficiency with environmental integrity under realistic industry, geographical, and regulatory conditions. Through a mathematical model, the study analyses varying national avoidance targets, illuminating the trade-offs confronting regulators, operators, and policymakers. Overall, the study provides an integrated techno-economic and governance-relevant assessment of CO₂ shipping networks, offering insights directly applicable to ongoing regulatory debates on cross-border transport, infrastructure planning, and long-term liability frameworks.

Transboundary CO₂ Shipping and CCS: Legal and Environmental Challenges

Levent Bilgili

Bursa Technical University, Turkey

The transboundary transportation of captured CO₂ by ship is gaining strategic importance as CCS deployment increasingly relies on offshore storage sites and cross-border infrastructure. Compared to fixed pipelines, ship-based CO₂ transport offers flexibility and scalability, enabling connections between dispersed emitters and suitable sub-seabed storage reservoirs. However, this growing role exposes significant legal and environmental challenges that existing regulatory frameworks address only partially. From a legal perspective, international instruments such as UNCLOS, the London Protocol, and selected IMO conventions provide a fragmented governance structure for cross-border CO₂ shipping. Although amendments to the London Protocol allow the export of CO₂ for sub-seabed sequestration under specific conditions, uncertainties persist regarding State consent procedures, permitting responsibilities, and liability allocation in the event of accidental releases during transport. Environmental challenges are closely linked to both operational risks and climate performance. The handling and maritime transport of liquefied CO₂ involve risks related to leakage, rapid phase change, and localised marine impacts, particularly in port areas and along major shipping routes. This paper argues that integrating well-to-wake life cycle assessment, as reflected in the IMO Life Cycle GHG Guidelines (MEPC.391(81)), into permitting and environmental impact procedures is essential to avoid burden-shifting across jurisdictions and to ensure that transboundary CO₂ shipping develops as a legally robust and environmentally credible component of global CCS governance.

Carbon Without a Category: Liquefied CO₂, Maritime Liability, and the Limits of Existing Pollution Regimes

Akrita Kaur and Rukhsaar Dhaliwal

Panjab University, Chandigarh, India

The rapid operationalisation of carbon capture and storage (CCS) at sea has outpaced the development of a coherent international liability framework governing the maritime transport and sub-seabed sequestration of captured CO₂. While existing international maritime liability conventions have evolved to address accidental pollution from oil and hazardous substances, their applicability to CCS remains uncertain. This paper argues that this uncertainty is not merely the product of regulatory gaps, but reflects a deeper structural misfit rooted in the ambiguous legal characterisation of liquefied CO₂ within international maritime law. Through a doctrinal analysis of UNCLOS, the London Convention and Protocol, and the principal maritime pollution liability regimes, the paper demonstrates that liquefied CO₂ defies existing regulatory categories. Neither conventionally understood as "cargo" nor as "pollution" in the traditional sense, CO₂ transported for permanent geological sequestration occupies a liminal legal position. This ambiguity undermines the triggering conditions and scope of application of existing liability regimes, rendering them ill-suited to address leakage, environmental harm, or long-term responsibility arising from CCS operations at sea.

03

Regulatory Frameworks & Legal Standardisation

15:15–16:55 · Alfa Laval Agri

Ingvild Ombudstvedt

Lena Østgaard

Carl Emil Meier

Henrik von Zweigbergk & Anna Månsson

Nicki Carnbrand Håkansson

International Subject Matter Experts Impact European Legal Framework for CCS Through Technical Standardisation

Ingvild Ombudstvedt

IOM Law, Norway

Technical standards are privately developed and owned documents, voluntary to use and with no direct link to legal frameworks. That said, industries such as oil and gas have a long tradition of technical standards being referenced in legal frameworks, relying on industry best practices rather than technical specifications embedded directly in regulations. A similar trend is emerging for CCS. Since 2011, the International Organization for Standardization has developed technical standards for CCS under Technical Committee (TC) 265, several of which are already referenced in legal frameworks and permitting procedures globally. In Norway, for example, the ISO/TC 265 standards for pipeline transportation (ISO 27913) and geological storage (ISO 27914) are referenced in the guidelines to the CO₂ Safety Regulation. In 2023, TC 474 was established under the European Committee for Standardization (CEN) to standardise CCS and CCU; it has already adopted ISO 27913 as a European standard in 2025 and is developing euro-centric standards, including CO₂ specification for transportation. The European Commission is expected to issue its first standardisation request to CEN/TC 474 in 2026, related to transportation and interoperability. This presentation highlights the role of technical experts and how standards developed under these committees are being used in legal frameworks for CCS, with a focus on CO₂ specifications for transport and storage.

Lessons from Norway's First-Mover CCS Projects: Regulatory Insights

Lena Østgaard

IOM Law, Norway

CCS is a key enabler for reaching climate goals. Today, industry awaits clearer regulatory guidance, while regulators depend on industry input and project development to inform the framework, creating a circular dependency that risks slowing progress. This presentation explores how industry and regulators work together to develop effective frameworks, emphasising the importance of a learning-by-doing approach. It examines three offshore Norwegian examples—Sleipner, Snøhvit, and Northern Lights—which have played a crucial role in helping regulators and industry identify gaps and barriers across the CCS value chain. These projects have directly influenced updates to Norway's CCS regulatory framework and further informed the EU's drafting of the 2009 CCS Directive. While Sleipner and Snøhvit serve to reduce and store CO₂ from petroleum production, the Longship project intends to receive CO₂ from a wide range of European emitters, thus testing not only national frameworks but also contributing to developments under the London Protocol enabling cross-border value chains. The presentation aims to show that projects can begin before regulatory frameworks are fully mature, to illustrate how frameworks need to be tested on real-world projects to become fit for purpose, and to highlight key cross-jurisdictional learnings from regulatory processes and demonstration projects.

The EU CCS Policy Landscape and State of Play

Carl Emil Meier

Bellona Europa, Belgium

Invited presentation — abstract not submitted.

Carbon Storage: Work Towards Simplification of Rules

Henrik von Zweigbergk and Anna Månsson

Geological Survey of Sweden (SGU), Sweden

The Geological Survey of Sweden has a governmental assignment on investigating the possibilities for simplification of rules regarding carbon storage. The purpose is to lessen the administrative burden, facilitate permitting, and fasten the permitting processes regarding carbon storage. The goal is to present relevant measures for simplification. The project has analysed barriers for carbon storage and will ground further work toward presenting measures for simplification. Examples of areas for further analysis include how to give carbon storage a stronger planning interest and national legal interest, increase knowledge about carbon storage, and how to effectively coordinate administrative processes and permit reviews. Legal barriers include the dumping prohibition in the Helsinki Convention, the ban in the CCS-directive of storing outside EU, and the ban in Swedish regulation of storing on land. The conclusion of SGU is that the main barrier is a combination of unclear regulation, knowledge gaps, and institutional challenges. There is missing an overarching national goal and strategy regarding CCS.

From Goals to Groundwork: SEA's Role in Making CCS Happen on the Road to Net Zero

Nicki Carnbrand Håkansson

Swedish Energy Agency, Sweden

This presentation outlines why CCS is essential for meeting Sweden's 2045 climate target and how national and EU legal frameworks shape the emerging CCS value chain, highlighting the Swedish Energy Agency's work to design policy instruments, address regulatory challenges, and support cross-border cooperation.



Hybrid Presentations

08:00–10:00 · Rättegångssalen

Morgane Querel

Rekhil Sukesan

Ashima Pury & Prapti Chaturvedi

Ayat-Allah Bouramdane

Mohammad Nizam Ashraf Khan & Bhavana Dhoundiyal

Bastiaan Ewoud Klerk

Howard Mwesigwa

The Importance of the Entry into Force of the HNS Convention

Morgane Querel

D-ICE Engineering / Lex Moana / Université de Nantes, France

The current international legal framework for the seaborne transboundary transportation of CO₂ fails to guarantee the protection of the marine environment. If a maritime casualty occurs, no international regulation currently provides a mechanism to establish liability, guarantee clean-up operations, environmental reinstatement, and victims' compensation. This role is fulfilled de facto by the 1976 Convention on Limitation of Liability for Maritime Claims (LLMC), which is not tailored for the transportation of CO₂.

Transboundary Maritime Transport of CO₂: Regulatory Vacuum and Legal Fragmentation

Rekhil Sukesan

Global Feeder Shipping LLC / Swansea University, UAE

The accelerated deployment of CCS as a climate mitigation strategy has elevated maritime transportation of liquefied CO₂ into a critical enabler of transboundary access to offshore sequestration sites. While ship-based CO₂ transport offers operational flexibility and circumvents the geographical and political constraints of pipeline infrastructure, it simultaneously exposes profound legal, environmental, and governance challenges that remain insufficiently resolved within existing international regulatory frameworks. This paper critically interrogates the legal and environmental risks associated with the transboundary shipment of captured CO₂ by sea, focusing on three interrelated problem areas: the contested legal classification of CO₂ as cargo or waste, the applicability and scope of marine dumping prohibitions, and the attribution of liability for leakage during carriage, transshipment, or incidental discharge. Central to the analysis is the IMO's London Protocol, particularly Article 6 and its 2009 amendment operationalised through the 2019 resolution. The paper demonstrates how fragmented ratification, uneven domestic implementation, and divergent interpretive practices continue to generate legal uncertainty for cross-border CO₂ shipping, and argues that existing instruments are ill-equipped to address the distinctive risk profile of liquefied CO₂ regarding cumulative impacts, fault attribution, and insurability.

Recalibrating Maritime Liability for Ship-Borne CCS: From Pollution Paradigms to Climate Infrastructure

Ashima Pury and Prapti Chaturvedi

Delhi Metropolitan Education (GGSIPU), India

The delivery in January 2026 of the world's first low-pressure liquid CO₂ carrier vessel by Capital Clean Energy Carriers Corp marks a significant development in the sustainable transport of gases by sea. Yet oceanic carriage liabilities continue to be anchored in the fragmented structures of conventional pollution regimes rather than in sound climate mitigation frameworks. Current instruments—including the Convention on Limitation of Liability for Maritime Claims and the HNS Convention—place a disproportionate burden on ship owners and fail to recognise CO₂ as a climate mitigation substance. The existing legal framework is characterised by significant lacunae, including the absence of differentiated liability for accidental leakage, inadequate treatment of transboundary harm from CO₂ releases, and insufficient compensation and insurance mechanisms. This is compounded by the absence of international jurisprudence and State practice in the intersection of climate and maritime law. This paper asks how maritime liability should be revised to govern ship-borne CCS operations in an environmentally protective and climate-mitigative manner, arguing for a fundamental shift toward treating ship-borne CO₂ transport as climate infrastructure rather than a residual variant of hazardous cargo carriage.

CCS in the Nordic Context: Balancing Climate Ambitions, Environmental Justice, and Socio-Ecological Impacts

Ayat-Allah Bouramdane

Independent Researcher, Rabat, Morocco

The urgency of mitigating climate change has intensified the need for rapid decarbonisation strategies across Europe. CCS has emerged as a potential technology to achieve EU, Swedish, and Nordic climate targets, yet its socio-ecological and environmental justice implications remain underexplored. This study investigates the role of CCS in meeting regional climate goals, addressing the central questions of how CCS can contribute to decarbonisation pathways in Sweden and the Nordic region and what the environmental justice and socio-ecological consequences of its deployment may be. Using a prospective modelling approach combining techno-economic assessments with spatial and social impact evaluations, the findings indicate that while CCS can significantly reduce industrial and power-sector emissions, the distribution of benefits and risks is uneven, with potential socio-ecological trade-offs including local environmental burdens, community acceptance challenges, and inequitable exposure to risks. The study underscores that CCS deployment must be guided by environmental justice principles and robust stakeholder engagement, and that policymakers should integrate CCS with complementary mitigation measures and monitor socio-environmental impacts to avoid exacerbating inequalities.

The Void Below: Bridging the Regulatory Gaps in Sub-Seabed Carbon Storage

Mohammad Nizam Ashraf Khan and Bhavana Dhoundiyal

IILM University, Greater Noida, India

As climate mitigation efforts intensify, sub-seabed carbon storage in Areas Beyond National Jurisdiction (ABNJ) emerges as a potentially significant solution for large-scale CO₂ sequestration. However, the current international legal framework presents substantial regulatory gaps that threaten the viability and safety of such operations. The existing patchwork of instruments—including UNCLOS, the London Protocol, and the emerging BBNJ Agreement—creates jurisdictional ambiguities, enforcement challenges, and liability uncertainties. The analysis advances three main arguments. First, the current regulatory void stems from the historical absence of carbon storage considerations in ocean governance frameworks, resulting in fragmented competencies between different treaty regimes. Second, the principle of the common heritage of mankind, while potentially applicable to sub-seabed resources, conflicts with carbon storage practices that require long-term territorial stability and clear property rights. Third, effective governance requires a new multilateral framework harmonising environmental impact assessment procedures, establishing transboundary liability mechanisms, and creating monitoring and verification protocols specific to ABNJ contexts. The paper proposes a pathway forward through institutional coordination between relevant international bodies and the development of technical standards adapted to the unique challenges of deep-sea carbon storage.

The ITLOS and ICJ Climate Opinions and CCS

Bastiaan Ewoud Klerk

UiT — The Arctic University of Norway (NCLOS), Norway

The recent climate advisory opinions of the International Tribunal for the Law of the Sea (ITLOS) and the International Court of Justice (ICJ) refine the content of States' obligations under international law, with indirect implications for carbon capture and storage (CCS). While neither opinion addresses CCS explicitly, both clarify due diligence duties under UNCLOS and the UNFCCC/Paris Agreement. They confirm that States must take all necessary measures, grounded in the best available science, to prevent, reduce, and control greenhouse gas emissions, and that this obligation is one of conduct requiring timely and effective action. These interpretations raise the expected standard of care, including stricter requirements for regulatory oversight, environmental impact assessment, and ongoing monitoring of mitigation activities. For CCS, this has two implications. First, it supports the characterization of CCS as a potentially necessary mitigation measure where it reflects best available technology in specific sectors. Second, it imposes more stringent procedural and substantive constraints on CCS deployment, particularly in transboundary and marine contexts governed by UNCLOS.

A Shield or a Double-Edged Sword? An Environmental and Social Justice Appraisal of Offshore Carbon Capture and Storage

Howard Mwesigwa

University of Edinburgh, United Kingdom

CCS has emerged as a prominent element in global decarbonisation pathways, particularly for hard-to-abate sectors. Increasingly, attention has turned to deployment in offshore and marine environments, framed by proponents as a climate mitigation “shield” capable of reconciling continued industrial activity with net-zero ambitions. Yet the expansion of CCS at sea raises profound environmental and social justice concerns. This paper critically appraises offshore CCS through an environmental and social justice lens, situating it within the evolving international legal architecture governing the marine environment, including UNCLOS, the London Protocol, and emerging climate governance norms. It interrogates whether existing regulatory frameworks adequately address long-term socio-ecological risks, transboundary harm, liability gaps, and cumulative impacts of sub-seabed carbon storage. Beyond environmental risk, the paper examines how offshore CCS projects may perpetuate or exacerbate structural inequalities by externalising environmental risks to coastal communities, marginalised fishing populations, and future generations, while delivering disproportionate benefits to major emitters in the Global North. The paper posits that, without robust legal safeguards and justice-centred governance, offshore CCS risks becoming a double-edged sword, and concludes by proposing normative principles and governance reforms to align deployment with environmental integrity, social justice, and the precautionary principle.



Keynote Address

11:00–11:20 · Tetra Laval

Niels Krabbe

University of Gothenburg, Sweden

Biological Carbon Sequestration vs CCS: Complementing MSY with Maximum Climate Mitigation (MCM)

Niels Krabbe

University of Gothenburg, Sweden

The oceans cover over 70 per cent of the planet's surface, are fundamental to human life, and play a crucial role in oxygen production, weather regulation, and the global carbon cycle. As the largest carbon sink in the world, they are estimated to have absorbed roughly 40 per cent of anthropogenic CO₂ emissions since the beginning of the industrial era. Yet contemporary climate strategies have largely focused on engineered solutions such as CCS, including offshore transport and sub-seabed geological storage, while the oceans' existing capacity to sequester carbon through biological processes has received comparatively limited legal attention. The importance of maintaining marine carbon sinks was formally acknowledged only with the adoption of the Paris Agreement and the United Nations Sustainable Development Goals in 2015, after which marine climate mitigation—often referred to as “blue carbon”—has increasingly been reflected in Nationally Determined Contributions and UNFCCC accounting frameworks. Recent scientific evidence demonstrates that the biological carbon pump, operated by marine organisms ranging from microbes to large vertebrates, binds and sequesters vast amounts of carbon at scales comparable to, and in some respects exceeding, those targeted by CCS. Unlike geological storage, these processes are self-maintaining, generate co-benefits for biodiversity and ecosystem resilience, and avoid many of the long-term liability and leakage risks associated with offshore CCS. Despite this, marine living resources continue to be managed under principles of optimum utilisation and Maximum Sustainable Yield (MSY), which prioritise extraction for human consumption and fail to account for the climate services provided by marine ecosystems. This paper argues that promoting biological carbon sequestration represents a more effective and potentially more cost-efficient climate mitigation pathway than expanding offshore CCS infrastructure, and develops the case for complementing MSY with Maximum Climate Mitigation (MCM) as a guiding objective for marine governance. While the BBNJ Agreement does not establish carbon sequestration as a central objective, its implementation provides a critical opportunity to prioritise ecosystem-based climate mitigation and embed the protection of biological carbon sinks within its mechanisms.

04

Civil Liability, Insurance & Risk

11:20–12:20 · Tetra Laval

Rosa Manzo

Romano Alessandro Bove

Albina Ladynenko

Climate Litigation, Due Diligence Obligations, and Public Interest Concerns in CCS Projects

Rosa Manzo

University of Bergen, Norway

The rapid expansion of climate litigation has significantly influenced the interpretation and enforcement of international climate obligations. In particular, courts and quasi-judicial bodies are increasingly engaging with the normative implications of the Paris Agreement when assessing States' duties of conduct in relation to climate mitigation strategies. This paper examines how recent decisions by international and regional courts contribute to the evolving understanding of climate-related due diligence obligations, with a specific focus on carbon capture and storage (CCS) projects. The central research question is whether, and to what extent, international adjudicatory practice is reshaping the legal standards applicable to CCS projects by embedding public interest considerations, human rights protection, and precautionary reasoning into the assessment of State responsibility. The paper argues that recent case law reflects a shift towards a more substantive conception of due diligence, requiring States not merely to adopt formal regulatory frameworks, but to actively evaluate the long-term compatibility of CCS projects with the objectives of the Paris Agreement. Drawing on selected international and regional decisions, the analysis highlights three emerging trends. First, courts increasingly frame climate obligations as dynamic and context-dependent, informed by scientific knowledge and evolving climate targets. Second, due diligence is interpreted as encompassing robust risk assessment, consideration of alternatives, and monitoring obligations throughout the lifecycle of CCS projects. Third, public interest concerns—particularly environmental integrity, intergenerational equity, and the protection of human rights—are becoming central to judicial scrutiny of climate mitigation technologies. The paper concludes that climate litigation is playing a critical role in clarifying the legal boundaries within which CCS projects may be pursued.

From Operator to State: Rethinking Civil Liability in Carbon Capture and Storage

Romano Alessandro Bove

University of Siena, Italy

Geological storage of CO₂, alongside emission reduction policies, represents a strategic tool for mitigating anthropogenic greenhouse gas emissions and achieving the objectives of the ecological transition. However, the development of investments in CCS critically depends on the existence of an adequate and efficient regulatory and civil liability framework. Significant challenges arise from the temporally extended nature of the risk, as potential harms to human health and marine ecosystems may materialise long after the operational phase has ended, placing considerable strain on traditional civil liability paradigms regarding the attribution of responsibility, the proof of causation, and the application of limitation periods. Directive 2009/31/EC seeks to address these concerns by introducing a mechanism for the transfer of liability to the State during the post-closure phase, subject to demonstration of permanent storage stability. Nonetheless, in many national legal systems a comprehensive and specific civil liability regime for CCS activities is still lacking. This paper argues for the design of a special liability regime that reallocates risk more efficiently while simultaneously promoting the development of CCS technologies. It proposes a dual-track system: strict operator liability during the phases of site selection, injection, and closure, followed by a transfer of responsibility to the State as custodian in the post-closure phase, coupled with mandatory insurance coverage to address the potentially severe consequences of CO₂ leakage.

Challenges for the Insurance Market in the Logistics Chain of Offshore CCS Projects

Albina Ladynenko

Universitat Rovira i Virgili, Tarragona, Spain

The development of offshore CCS projects—from the beginning of the logistics chain in the port through to maritime transport—poses significant legal and regulatory complexities, particularly with regard to the allocation and insurability of environmental and operational risks among all operators involved. Traditional marine and energy insurance regimes may not fully accommodate the specific characteristics of long-term CO₂ storage, its temporary storage in ports, and its maritime transport between facilities. This presentation examines the challenges that the insurance market faces in addressing these emerging needs and the gaps that persist in both coverage and regulation as a mechanism for risk transfer safeguarding the economic viability of the business in the event of unforeseen loss. It considers the responses and adaptations required in the areas of construction and installation insurance, civil liability, environmental coverage, and long-term post-operational liability, as well as the guarantees and mechanisms necessary to ensure adequate risk transfer and financial security throughout the lifecycle of storage sites, particularly for risks extending beyond the time horizons of insurers.

05

National Legislation & Cross-Border Governance

13:20–15:20 · Tetra Laval

Daopu Somoni

Patrick Seroogy

Clément Lasselin

Bára A. Alexandersdóttir

Josep Lluís Díez i Besora

Michał Krzykowski & Bartosz Prusik

Emerging Regulatory Strategies Across the North Atlantic for CCS Technology Deployment

Daopu Somoni

University of Aberdeen, United Kingdom

Carbon Capture and Storage (CCS) technology is one of the solutions identified and being deployed to address decarbonisation of CO₂ emissions in the energy transition process. The technology is presently predominantly deployed in advanced economies such as Norway, Canada, Australia, United Kingdom, and United States, while similar efforts are being advanced in emerging jurisdictions such as South Africa and Nigeria. An examination of these advanced economies reveals the adoption of various regulatory strategies—'value chain regulatory strategy', 'whole-of-government regulatory strategy', and multi-tier governance strategy (predominant in Federal systems). All these strategies are embedded in an overarching 'command-partnership' regulatory approach. Therefore, given that emerging economies are developing CCS deployment policies and regulatory frameworks, verifiable lessons from CCS regulatory strategies in advanced jurisdictions are relevant in deepening and expanding seamless global CCS deployment.

Balancing Climate Goals and Environmental Risks Through the Precautionary Principle: A Case Study of the Baltic Sea

Patrick Seroogy

George Washington University Law School, United States

CCS represents a serious measure for achieving climate neutrality. Due to limited onshore capacity, countries are considering sub-seabed sequestration of CO₂ in geological formations that promise secure, long-term storage. However, this method presents challenges for marine ecosystems, including CO₂ leakage, long-term monitoring, and liability. In the Baltic Sea, coastal states have begun assessing sub-seabed sites for carbon sequestration and are considering redefining “dumping” under the Helsinki Convention to permit such storage. The Baltic’s unique properties—shallow depth, brackish water, and hypoxic seafloor—make it particularly vulnerable to environmental degradation. This paper argues that any Baltic Sea CCS project should require strict application of the precautionary principle as codified in the Helsinki Convention. Central to this application is the establishment of cross-boundary scientific consensus that a given CCS operation does not present serious risk of environmental harm before it may proceed. The Helsinki Commission could oversee this process and establish reasonable metrics, with consensus requiring multinational agreement. Such strong requirements are vital because scientific advice functioning solely as guidance can be sidestepped by short-sighted decision-making.

Marine CCS and Basaltic Rocks: Inaccessibility and Promises

Clément Lasselin

Lund University (Faculty of Engineering), Sweden

Carbfix's proposed Coda Terminal is a novel cross-border carbon capture and storage (CCS) project that aims to import up to three million tonnes of CO₂ annually to Iceland for onshore mineral storage. While distinct projects, perceptions of the now-defunct Running Tide project—a marine carbon dioxide removal (mCDR) startup aimed to sequester carbon in the ocean through ocean alkalinity enhancement and algae production—has affected the regulatory and political environment for CCS projects like Coda. Both Carbfix and Running Tide were first-of-their-kind projects in Iceland, requiring the development of bespoke regulatory and permitting frameworks. Carbfix has played a key role in fostering the development of cross-border CO₂ transport and the place of mineralization in EU ETS schemes, but proper governance and policy mechanisms for Iceland to utilize CCUS to help contribute to its greenhouse gas emission reduction targets are lacking. Drawing on anthropological fieldwork conducted in Iceland and abroad with regulators, policymakers, and other actors involved in cross-border carbon transport and storage at sea, this paper examines how these two projects have shaped perceptions of environmental governance and political integrity in emerging CCS and CDR regimes.

Carbon Storage and Cross-Border Transport: Domestic Permitting Challenges in Iceland

Bára A. Alexandersdóttir

Reykjavík University, Iceland

CCS is increasingly recognised as a necessary component of global climate mitigation strategies. This paper examines how Iceland's domestic legal and administrative framework addresses the permitting of ship-transported CO₂ for permanent storage, with particular focus on the legal uncertainty surrounding cross-border transport. The analysis situates CCS within Iceland's climate policy framework, including its statutory commitment under the Climate Act No. 70/2012 to achieve carbon neutrality by 2040. Iceland's extensive basalt formations, extending both onshore and beneath the seabed, enable rapid and permanent mineralisation of injected CO₂, significantly reducing long-term leakage risk compared to conventional sedimentary storage. Such geological advantages have attracted international interest, yet CCS projects in Iceland face considerable legal uncertainty. The Coda Terminal project by Carbfix represents a pivotal case for carbon storage governance involving ship transport. Supported by a €3.9 million EU grant, the project was designed to receive CO₂ shipped from European emitters for onshore mineralisation. Its permitting process intersected domestic administrative procedures with international obligations under the London Protocol and the OSPAR Convention. The project also illustrates the relevance of the Aarhus Convention, as public concerns about Iceland becoming a repository for foreign CO₂ shaped the permitting process and contributed to administrative delays that ultimately halted the project in Hafnarfjörður. This paper situates Iceland's domestic framework within international climate law developments and offers a notable case study for understanding how domestic permitting frameworks engage with emerging and often fragmented international governance.

Adaptability Challenges of Ports for the Viability of Offshore CCS Projects

Josep Lluís Díez i Besora

Universitat Rovira i Virgili / Tarragona Port Authority, Spain

Offshore CCS projects are increasingly positioned as a strategic element in energy transition pathways, particularly for hard-to-abate industrial sectors. In this context, ports emerge as critical energy and logistics hubs, acting as interfaces between onshore carbon capture facilities, transport networks, and offshore storage sites. The viability of offshore CCS projects is therefore closely linked to the capacity of ports to adapt their operational models, infrastructure, and governance frameworks to the specific requirements of CO₂ handling and transport. This paper analyses the role of port adaptability in enabling offshore CCS deployment, with a specific focus on operational changes and concession regime implications. Ports must integrate new energy-related functions—including the reception, temporary storage, conditioning, and transfer of CO₂—requiring specialised infrastructure, enhanced safety and environmental management procedures, and modifications to terminal operations and intermodal connectivity. From an institutional and regulatory perspective, current port concession regimes present a significant challenge: traditional concession models are typically structured around established cargo flows and relatively short investment cycles, whereas CCS projects are characterised by long development timelines, high capital intensity, and substantial regulatory and market uncertainty. The paper argues that ports must evolve from conventional transport nodes into adaptable energy hubs, requiring coordinated planning between port authorities, energy stakeholders, and regulators, as well as the development of flexible concession frameworks aligned with the long-term nature of offshore CCS projects.

In Search of Effective Legal Solutions for CCS: Quo Vadis, Polish Legislator?

Michał Krzykowski and Bartosz Prusik

University of Warmia and Mazury in Olsztyn, Poland

Despite possessing one of Europe's largest CO₂ storage capacities, estimated at 12 to 15 gigatonnes, Poland has historically struggled to establish a functional CCS sector. For over a decade, national regulations remained practically dormant, strictly limiting CCS exclusively to demonstration projects and prohibiting commercial utilisation. By 2024, not a single entity had applied for a licence to transmit or store CO₂. However, recent shifts in European energy policy, driven by the European Green Deal and the Net Zero Industry Act, have catalysed a legislative change. The pivotal 2024 amendment to the Polish Geological and Mining Law Act abolished the demonstration-only restriction, enabling commercial deployment. Furthermore, it introduced regulatory exemptions for small-scale storage (under 100 kilotonnes) and streamlined administrative procedures by replacing rigid concession requirements with less formal geological works approvals. This legal evolution aligns with the progressive integration of CCS technologies into Poland's energy policy and national decarbonisation strategies, as illustrated by the planning of cross-border infrastructure such as the ECO2CEE Project of Common Interest. This paper critically examines whether these recent legislative changes are sufficient to stimulate robust CCS development. While the 2024 reforms constitute a significant turning point, the sector remains hindered by regulatory fragmentation, including the lack of a single dedicated legal act and competent authority.